BioTest: Metamorphic Testing for Textual VCF and SAM Parsers

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***Abstract*—**
Bioinformatics file-format parsers determine whether a variant call, alignment flag, or sample annotation reaches downstream analysis intact. Pipeline-level benchmarking efforts [1] assume parsers are sound, yet the textual VCF and SAM specifications leave many record-level semantics to de facto consensus, so testers operate without an external oracle [2]. We present BioTest, a metamorphic testing pipeline that replaces the human MR-identification step, the dominant cost in metamorphic-testing adoption [3], with a large language model (`deepseek-v4-flash`) grounded in the published specifications via retrieval-augmented prompting. The LLM mines identity-preserving relations once per format. Inputs flow through a thirteen-rank corpus stack and are graded by multi-runner consensus across $N{\geq}3$ independent implementations. The per-SUT user cost is the parse-and-emit shim that mainstream coverage-guided fuzzers already require, with a small canonicalization extension that emits canonical JSON. Across six $(\mathrm{SUT}, \mathrm{format})$ configurations spanning four parser languages, BioTest sits within a single-digit mutation-adequacy deficit on five configurations, exceeds libFuzzer on seqan3/SAM, and trails Atheris by 32.5 pp on byte-content-lenient biopython/SAM. It attains the highest line coverage on every VCF SUT, and on a real-bug benchmark of 32 historical parser bugs [4] detects 10, spanning htsjdk (Java), vcfpy (Python), and noodles (Rust), where every crash-only input fuzzer in the slate detects zero.

***Index Terms*—metamorphic testing, mutation testing, fuzz testing, bioinformatics, parser testing, oracle problem, software testing**

I. Introduction

A misread alignment flag, a 1-based coordinate parsed as 0-based, or a silent record drop in a Variant Call Format (VCF) file propagates from the parser to every downstream tool that consumes its output. Independent variant-calling pipelines disagree substantially on the same input data [5], and the discrepancy has motivated sustained investment in pipeline-level benchmark infrastructure such as Genome in a Bottle [6] and the GA4GH best-practice protocol [1]. These efforts grade the joint behaviour of caller and parser under the assumption that parsers are sound. Whether they are is rarely tested directly. Concrete consequences are easy to find in upstream issue trackers: htsjdk SAM regressions that accepted reference names with delimiters disallowed by SAM 1.6 and that over-strictly rejected reads with template lengths above 2^{29}, and a vcfpy regression that silently accepted symbolic ALT alleles VCF restricts to gVCF mode. Such regressions corrupt records that downstream callers treat as authoritative, and the error propagates to every downstream analysis. The textual specifications fix the on-disk grammar but leave many record-level semantics to de facto consensus, so testers operate without an external oracle [2].

Metamorphic testing replaces "what should the output be?" with relations on related outputs [3], [7]. File-format parsers fit cleanly: identity-preserving transforms over the spec-defined grammar yield equivalence relations derivable from the specification rather than from the parser source. The dominant cost is the relations themselves. The field's open-challenge survey [3] catalogues MR identification as the central problem, and existing studies report relations specified per SUT by human experts in the SUT's host language [8]. The bioinformatics ecosystem maintains parsers for VCF and SAM in four host languages simultaneously (htsjdk in Java, biopython and vcfpy in Python, noodles in Rust, seqan3 in C++), so a hand-authored per-SUT pipeline does not scale. The strongest competing approaches do not scale either: coverage-guided fuzzers require bytecode rewriting, sanitizer-instrumented builds, or fork-server harnessing on top of the parse-and-emit shim every fuzz tool needs.

We describe BioTest, a metamorphic testing pipeline that replaces the human MR-identification step with an LLM-driven, spec-grounded mining stage. The published specification is indexed for retrieval-augmented prompting [9]. An LLM (`deepseek-v4-flash`) proposes identity-preserving transforms grounded in the spec rule each transform claims to preserve, validated against a seed corpus before admission. Inputs flow through a thirteen-rank stack (Ranks 1–7 metamorphic, Ranks 8–13 augmentation), and the oracle is multi-runner consensus across $N{\geq}3$ independent parsers. The user's per-SUT effort is a parse-and-emit shim that emits a canonical-JSON record set. BioTest adds no per-SUT MR, oracle predicate, seed-corpus curation, or instrumentation on top. Figure 1 gives the visual overview.

We evaluate BioTest on six $(\mathrm{SUT}, \mathrm{format})$ configurations against five coverage-guided fuzzers and a Pure-Random floor across three axes: mutation adequacy, end-of-run line coverage, and a real-bug benchmark [4] of 32 verified parser bugs (a tool detects a bug iff some input it produces triggers the buggy version but not the patched version). BioTest is competitive with the strongest coverage-guided fuzzer on text-rich parsers and trails on byte-content-lenient ones, consistent with the empirical pattern of Klees et al. [10] and Böhme et al. [11]. On the bug benchmark it detects 10 of 32 parser bugs across three host languages (Java/htsjdk, Python/vcfpy,

Rust/noodles), where every crash-only input fuzzer in the slate detects zero. Detection is end-to-end audited under a prefix / post-fix differential predicate in the ground-truth fuzzing style of Magma [4], on inputs the tool's adapter produced during the per-cell budget rather than on the manifest's canonical proof-of-vulnerability files alone.

a) Contributions.:

(i) BIOTEST, a metamorphic testing pipeline for textual VCF and SAM parsers in which an LLM mines spec-grounded MRs (rather than per-SUT hand authoring) under RAG retrieval and seed-corpus validation, with parser behaviour graded by a multi-runner consensus oracle.

(ii) A thirteen-rank corpus stack that combines grammar-driven metamorphic transforms with structural and lenient-byte augmentation, plus empirical characterisation of when the augmentation ranks fail.

(iii) A three-axis evaluation (mutation adequacy, line coverage, 32-bug real-bug benchmark) under matched protocols that maps where grammar-driven generation matches, leads, and trails coverage-guided fuzzing under the bounded per-SUT effort envelope (§V).

b) Roadmap.: §II surveys prior work in metamorphic testing, coverage-guided fuzzing, mutation testing, and bioinformatics parser correctness. §III and §IV present the design and implementation of BIOTEST. §V reports the three-axis evaluation, and §VI discusses threats to validity. §VII concludes.

II. RELATED WORK

BIOTEST sits at the intersection of metamorphic testing (supplying the oracle), coverage-guided greybox fuzzing (the strongest baselines), and mutation testing (the adequacy criterion). Our emphasis is on positioning rather than survey recapitulation.

A. Metamorphic Testing

Metamorphic testing was introduced by Chen et al. [7] as a partial answer to the oracle problem [2], [12]. Rather than asserting an absolute output, one asserts a metamorphic relation constraining how outputs change under structured input perturbations. Segura et al. [8] survey empirical adoption and Chen et al. [3] catalogue the open challenges, of which MR identification is the dominant cost in the technique's adoption.

Several lines of work attack MR identification by mining relations from existing artefacts. MeMo [13] extracts MRs from natural-language phrases in Javadoc comments, and MR-Scout [14] synthesises MRs from user-written test cases. BIOTEST mines from a different source, the published format *specification* itself. An LLM, grounded in the spec text via retrieval-augmented prompting and gated by a seed-corpus validator, proposes identity-preserving transforms that instantiate a parametric round-trip relation ($out(parse(x)) \equiv out(parse(T(x))))$. The novelty we claim is the combination of spec-grounded LLM-driven MR mining with a SUT-agnostic generator and consensus oracle, not any individual component.

B. Coverage-Guided Greybox Fuzzing

The in-process baselines we compare against typically find 2 to $5\times$ more bugs than blackbox baselines on a 24-hour budget [10], an advantage Böhme et al. [11] explain information-theoretically. Grammar-aware fuzzers (Zest [15], Nautilus [16], Superion [17], Smart Greybox Fuzzing [18], Fuzz4All [19]) recover structural fluency but require SUT-specific harness and build work that exceeds C2.

C. Mutation Testing

We use mutation testing as an adequacy proxy. Just et al. [20] establish that mutation kills correlate with real-fault detection at $r\approx0.7$ on Defects4J, and Andrews et al. [21] earlier showed mutants are an acceptable substitute for hand-seeded faults under standard operator sets. Papadakis et al. [22] survey the field and catalogue the equivalent-mutant problem. We adopt Vikram et al. [23] kill-feedback corpus selection at the pre-grading minimisation step. We complement the proxy with the real-bug benchmark of §V-C.

D. Bioinformatics Pipeline and Variant-Caller Testing

The bioinformatics community has built benchmarking infrastructure around *variant-caller correctness*. Genome in a Bottle [6] provides gold-standard variant calls, and the GA4GH best-practice protocol [1] (`hap.py`, `vcfeval`) controls for representational equivalence between a candidate VCF and the truth set. O'Rawe et al. [5] document substantial inter-pipeline disagreement on the same input data. These efforts grade the joint behaviour of caller, reference, and downstream parsing under the assumption that VCF and SAM parsers are sound. BIOTEST attacks the parser-correctness layer directly. A metamorphic violation, parser disagreement, or silent record drop that BIOTEST surfaces is a defect in a component that GIAB-style benchmarks assume to be sound, and the two efforts are therefore complementary.

III. DESIGN

BIOTEST treats a parser as an opaque function from bytes to a structured record set and exercises it via spec-derived metamorphic relations [3], [7], [8], multi-runner consensus oracles, and coverage-steered corpus augmentation. The framework remains useful across heterogeneous parser implementations of the same format without per-SUT engineering effort beyond the parse-and-emit shim that coverage-guided fuzzers already require. VCF [24] and SAM [25] are partially context-sensitive textual formats: header declarations constrain record-level types, and they admit multiple semantically-equivalent surface forms (tag order, integer encodings, mandatory-vs-default values) that make them amenable to metamorphic testing.

A. Design Constraints

BIOTEST is built on three constraints. **C1: SUT-agnostic core.** The framework contains no per-SUT logic in any phase. **C2: Per-SUT user effort bounded by what coverage-guided fuzzers already require.** Onboarding a parser is a thin *harness* that calls the parser and emits a canonical-JSON record set, plus a ∼50-line Python *runner* subclass that wraps the harness for the orchestrator. The harness conforms to a fixed per-format canonical schema (typed fields, deterministic ordering, 1-based coordinates). BIOTEST adds *no other per-SUT artefact*: no metamorphic relation, oracle predicate, seed corpus, instrumented build, bytecode-rewriting agent, or fork-server harness. **C3: Semantic-digest oracle.** A violation is declared when the observable parse outcome (parse-success flag, canonical-JSON digest, exception class) diverges, with no per-SUT assertion code. The oracle's intermediate strength forfeits coverage-feedback's bug-finding advantage on byte-content-lenient parsers [10], [11] in exchange for portability across four parser languages.

B. Pipeline

The framework runs as five phases (Figure 1). Phases A and B run once per format, and phases C, D, and E run per iteration on the SUT pool. **Phase A** chunks the published specification and embeds the chunks into a vector store for retrieval-augmented prompting [9], parametric over format and spec version. **Phase B** prompts an LLM to propose candidate metamorphic relations $out(parse(x)) \equiv out(parse(T(x)))$, each grounded in spec rules retrieved from the Phase A store and validated against a reference parser on a seed corpus before admission. **Phase C** routes each input x and its transform $T(x)$ through every configured parser runner. Outputs are canonicalised to JSON and graded by the consensus oracle of §III-D, which layers metamorphic equivalence on top of differential agreement [26]. **Phase D** reduces coverage telemetry to per-class blind-spot tickets and folds the highest-yield uncovered classes back into Phase B's prompt. **Phase E** content-hashes inputs accepted by at least one runner and runs the augmentation operators of §III-C (Ranks 8–13).

C. Corpus Stack

A pure metamorphic test loop ceilings at modest line coverage on file-format parsers. We therefore widen the corpus along two tiers ordered by decreasing semantic preservation. Ranks 1–7 are semantics-preserving metamorphic transforms whose outputs the consensus oracle scores under the metamorphic predicate. Ranks 8–13 trade strict semantic equivalence for branch and value diversity, and their outputs feed mutation grading and the differential-only oracle, which does not require metamorphic validity to declare a kill.

- **Ranks 1–7 (metamorphic).** LLM-synthesised seeds (R1), upstream-derived corpora (R2), spec-rule-targeted malformed mutators (R3), Hypothesis-targeted property-based generation (R4), API-query relations $P(parse(x)) \equiv P(parse(T(x)))$ over reflection-discovered scalar getters (R5), opt-in LLM-synthesised relations (R6), and theme-restricted spec-invariant registries (R7).
- **R8 corpus keeper.** Persists every transformed input accepted by at least one runner, deduplicated by canonical AST-hash.
- **R9 value diversifier.** Perturbs spec-bounded numerical fields (POS, QUAL, MAPQ, AF, DP, TLEN) within bounds, gated by a canonical-normaliser validity check.
- **R10 byte fuzzer.** Validity-gated bit-flip, byte-substitution, byte-insert, and byte-delete operators on non-header positions.
- **R11 boundary-value generator.** Boundary Value Analysis [12] on spec-bounded numerical fields.
- **R12 structural variant generator.** Spec-derived enumerator emitting branch-diverse inputs (CIGAR-shape mixes, tag-type permutations, multi-sample records, varied FILTER/INFO/FORMAT compositions).
- **R13 lenient byte fuzzer.** Weak-gate byte-level mutator retaining files with at least one line. Not metamorphic, consumed only by mutation-adequacy grading.

D. Consensus Oracle

Phase C parses each input with $N \geq 3$ SUTs drawn from the runner pool. The VCF pool is htsjdk, vcfpy, noodles, and pysam. The SAM pool is htsjdk, biopython, seqan3, and pysam. When the SUT under grading is one of these runners, the remaining three voters supply the consensus. Each runner emits a parsed record set canonicalised to JSON and hashed, alongside an exception class on failure. The oracle classifies the input under three layered predicates. (i) Differential consensus: if a strict majority of runners agrees on a digest $\sigma^\star$ and runner i disagrees, runner i is implicated. (ii) Metamorphic consensus: if $\sigma^\star_x = \sigma^\star_{T(x)}$ but $\sigma^x_i \neq \sigma^{T(x)}_i$, runner i is implicated. (iii) Relation quarantine: if the consensus disagrees on x versus $T(x)$, the relation has misclassified what the spec says is semantics-preserving. The choice $N \geq 3$ is the minimum that admits a strict majority and distinguishes "one runner is wrong" from "the spec is ambiguous" [2].

E. Cross-Tool Fairness Recipe

§V grades BIOTEST against coverage-guided fuzzers under four identical conditions. Each pair uses the same mutation engine (PIT [27] for htsjdk, mutmut [28] for vcfpy and biopython, cargo-mutants [29] for noodles, a DIY mull-style AOR/ROR/LOR operator set [20], [30] for seqan3), the same target-class scope, the same kill predicate (parse-success flip, canonical-JSON divergence, or crash-class flip) scored as $killed/reachable$ [22], [27], and the same wall-clock budget per pair across n reps. Only the corpus differs.

Wall-clock parity is not iteration-cost parity. In-process fuzzers iterate faster than fork-server fuzzers, and the imbalance shows up on seqan3/SAM as a 17 pp gap between AFL++ [31] and libFuzzer [32] despite identical mutants. Every corpus is trimmed to ≤ 200 files by an AFL-cmin-style minimiser using per-file kill-cardinality on Python SUTs [23]

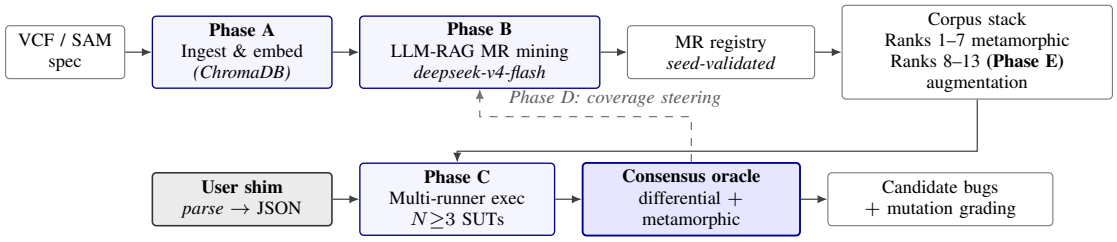

Fig. 1. BIOTEST pipeline. Phases A and B run once per format, and phases C, D, and E run per iteration. The dashed arrow marks Phase D coverage feedback into Phase B. The user's only per-SUT artefact is the parse-and-emit shim, the same artefact in-process coverage-guided fuzzers require.

and an SUT-agnostic outcome-fingerprint elsewhere. The selector adds $+1$ to $+2$ pp on pairs where it actively trims. As an uninformed-fuzzing reference [10] we add a Pure-Random byte emitter under identical conditions and report it as a per-pair floor.

IV. IMPLEMENTATION

BIOTEST is a Python orchestrator. Per-language SUT runners delegate parser invocation to the SUT's native toolchain, and the five phases of §III are realised concretely as follows.

a) Specification ingest (Phase A).: The published format specification is segmented at rule and field-definition boundaries, embedded with a sentence-transformer encoder, and indexed in a ChromaDB vector store. The same store serves every SUT for the format and is rebuilt only when the spec version changes.

b) LLM mining (Phase B).: The runs reported here use `deepseek-v4-flash` via the DeepSeek API. The orchestrator is provider-agnostic. The prompt retrieves the top-ranked spec chunks by cosine similarity to the rule the proposal must cite [9] and constrains the model to a whitelist of $\sim$ 20 atomic transforms registered in the framework. Each candidate MR traverses three sequential gates. The whitelist gate rejects unrecognised transforms. The hash gate collapses candidates whose $(format, scope, transform_steps)$ tuple matches an already-admitted MR, so LLM nondeterminism does not inflate the registry. The seed-corpus gate validates the round-trip equality on a hand-curated reference corpus before admission, and is the load-bearing rejection cause across recorded snapshots. The first targeted mining run on the `ordering_invariance` intent for VCF admitted three enforced MRs out of seven candidate themes, and the per-pair enforced count in the headline runs ranges from 3 for SAM to 10 for VCF. Figure 2 traces the pipeline through one such accepted MR with real artefacts from the spec chunk to the validation outcome.

c) Runner architecture (Phases C, D).: The runner subprocesses the SUT's harness, parses the canonical-JSON digest, and exposes the result through a uniform interface. Java harnesses run on a per-test JVM, Python harnesses import the parser directly, Rust harnesses run via `cargo test` against a generated crate, and C++ harnesses are compiled once as a gcov-instrumented binary. Coverage is collected with JaCoCo for Java, `coverage.py` for Python, `cargo-llvm-cov` for Rust, and `gcovr` for C++.

d) Augmenters (Phase E).: Rank 12 and Rank 13 ship as stand-alone generators that are not LLM-mined. Rank 12 outputs feed both the metamorphic oracle and mutation grading. Rank 13 outputs feed only the latter, since Rank 13 is intentionally not metamorphic.

e) Reproducibility.: The orchestrator, SUT source pins, mutation engines, coverage tools, and baseline fuzzers ship as a single Docker image on Ubuntu 22.04. The C++ mutation engine is a re-implemented default-operator set rather than upstream Mull [30] due to a glibc/Ubuntu version conflict. The artefact is deposited as an anonymised Zenodo record (DOI placeholder for double-blind review, de-anonymised at camera-ready) containing the Docker image digest, the source-tree snapshot, the MR-curation log, and the per-pair raw kill counts and bug-bench records. We commit to the venue's artefact-evaluation track.

V. EVALUATION

We evaluate BIOTEST along three complementary axes. §V-A reports mutation adequacy of the corpus produced by BIOTEST relative to corpora produced by mainstream coverage-guided fuzzers under the matched-engine fairness recipe of §III-E. §V-B reports line coverage on the same scope and SUT slate. §V-C reports a real-bug benchmark of 32 verified historical parser bugs. Mutation kills correlate with real-fault detection at $r{\approx}0.7$ [20], so the three axes together provide construct triangulation. Per-pair sample sizes are reported in Table I.

The evaluation spans six $(\mathrm{SUT}, \mathrm{format})$ pairs: htsjdk/VCF, htsjdk/SAM (Java), vcfpy/VCF, biopython/SAM (Python), noodles/VCF (Rust), and seqan3/SAM (C++). All runs execute inside the reproducibility container described in §IV.

A. Mutation Adequacy

a) Setup.: We grade the corpus produced by BIOTEST under the fairness recipe of §III-E. Six $(\mathrm{SUT}, \mathrm{format})$ pairs span four parser languages and four mutation engines. The four close-margin pairs (htsjdk/VCF, htsjdk/SAM, vcfpy/VCF, noodles/VCF) run at $n{=}10$ reps on the BIOTEST side after a power-driven boost from $n{=}4$. Biopython and seqan3 retain $n{=}4$ where samples are deterministic ($\sigma{=}0.00$) or operate at the protocol's budget-termination point. We report mean $\pm$ sample std. For biopython, the Atheris baseline terminates at the 262nd of 523 generated mutants under shuffle seed 42, and

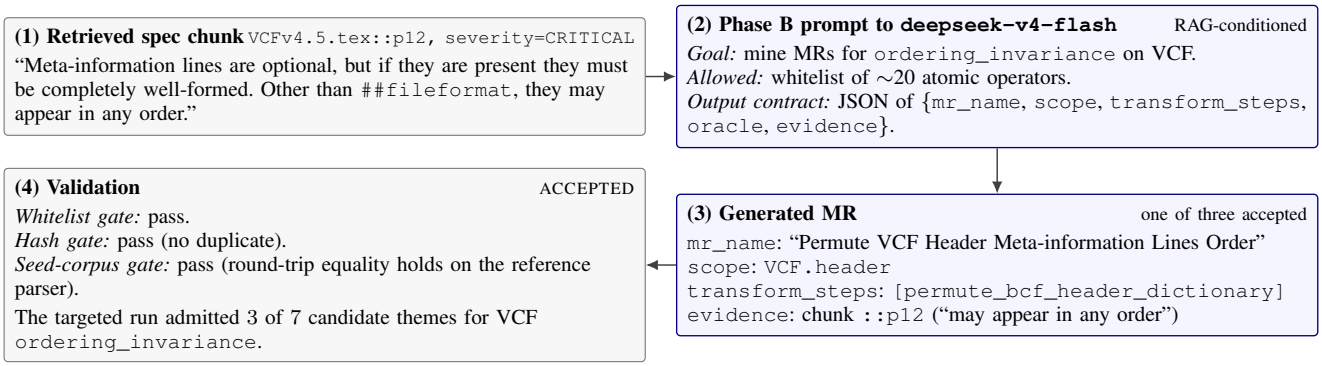

Fig. 2. LLM I/O traced through one accepted metamorphic relation. Panels 2 and 3 are LLM-touched (blue), panels 1 and 4 are framework data (grey). Every panel reproduces an artefact committed in the BioTest source tree.

we cap BIOTEST at the same 262 mutants for protocol parity, additionally reporting the full-population result.

b) Significance testing.: On the four close-margin pairs we use $n_{\mathrm{BT}}{=}10$ and the achieved baseline n shown in Table I. We compute exact two-sided Mann–Whitney U by full enumeration of the $\binom{n_1+n_2}{n_1}$ rank arrangements (no normal approximation), Vargha–Delaney $\hat{A}_{12}$ [36], a non-parametric effect size equal to the probability that a randomly drawn BIOTEST sample exceeds a randomly drawn baseline sample (0.5 indicates no effect, values approaching 0 indicate BIOTEST is consistently below, and values approaching 1 indicate BIOTEST is consistently above), with conventional small/medium/large thresholds at $0.56/0.64/0.71$, and Holm–Bonferroni correction at $k{=}4$. All four pairs clear family-wise $\alpha{=}0.05$, with the strongest signal at htsjdk/VCF (Holm $p{=}4.0\times10^{-5}$) and the weakest at vcfpy/VCF (Holm $p{=}3.7\times10^{-2}$). The descriptive ranking (BIOTEST below the strongest baseline on the dominant majority of rep-pairings) is now an inferential conclusion at family-wise $\alpha{=}0.05$ [37], not a small-sample observation.

c) Biopython is where coverage-guided fuzzing genuinely wins.: The largest signal is the -32.52 pp deficit on biopython, under the matched-budget protocol that caps both BIOTEST and Atheris at the first 262 generated mutants. Biopython's parser applies a strict structural validator but accepts byte-level variation within field values, and Atheris accumulates files that pass the validator and trip rare error paths. Enabling our Rank-12 structural and Rank-13 lenient-byte-fuzz augmentation does not move the mean ($25.48{\pm}0.79$ vs. un-augmented $25.57{\pm}0.00$, $\hat{A}_{12}{=}0.50$). This is the empirical pattern Klees et al. and Böhme et al. [10], [11] report on byte-content-lenient parsers. Closing the gap requires per-SUT bytecode instrumentation, outside our C2 envelope.

d) Four close-margin configurations.: On htsjdk/VCF, htsjdk/SAM, vcfpy/VCF, and noodles/VCF the deficit to the strongest baseline is in the 1.5 to 4.2 pp range, and BIOTEST is below the baseline on the dominant majority of $n_{\mathrm{BT}}{\times}n_{\mathrm{base}}$ rep-pairings ($\hat{A}_{12}\in\{0.000, 0.083, 0.111, 0.183\}$, all in the "large" magnitude band). We do not claim "match": Klees et al. [10] report a 2–5× advantage for coverage-guided fuzzers, and we measure how close grammar-driven generation can come under our design constraints. On vcfpy the absolute kill count remains competitive: BIOTEST kills $851.5{\pm}1.4$ of 992 reachable, Atheris kills $832.0{\pm}30.6$ of $952.4{\pm}50.5$ reachable.

e) seqan3: the consensus oracle's retained corpus matters.: On seqan3/SAM BIOTEST exceeds libFuzzer [32] by $+6.87$ pp on the means, with both samples deterministic ($\sigma{=}0.00$) under repeated runs against the same corpus. We report only the mean difference and decline to attach an $\hat{A}_{12}$ value, which on a pair of scalar means is uninformative beyond the sign. The mechanism is a corpus-retention difference under identical engine-side kill semantics. BIOTEST's spec-driven generator constructs inputs reaching distinct typed-exception classes (CIGAR-shape, tag-type, and flag-bit boundaries via Rank 12), whereas libFuzzer's coverage-guided byte mutator preferentially retains crashes and discards inputs that merely produce a typed exception. Mutations flipping one typed-exception class to another register as kills only when the corpus contains an input reaching the original class [2].

Construct- and internal-validity threats specific to this measurement are detailed in §VI.

B. Code Coverage

a) Setup.: We complement mutation-adequacy with a code-coverage measurement on the same six $(\mathrm{SUT}, \mathrm{format})$ pairs, scoped by the same target-class filter the mutation engine uses (§III-E) so the two metrics are graded on the same SUT surface. Line coverage (the fraction of source lines in each parser executed at least once during a tool's campaign) is collected by each language's standard coverage instrumentation (specific backends are listed in §IV). We report final-attainment line coverage, the end-of-campaign value each tool reaches inside its scope, because BIOTEST's Phase-D feedback loop (Figure 1) terminates on its own stop-criteria rather than at a fixed wall-clock tick. Within each row of Table II the scope filter and instrumentation backend are identical across columns. Line percentages are directly comparable as final attainment.

b) Headline observations and take-away.: On the three VCF SUTs BIOTEST attains the highest mean final line

TABLE I

MUTATION-ADEQUACY MEAN ± SAMPLE STD (%) ACROSS THE SIX (SUT, format) PAIRS. IDENTICAL MUTATION ENGINES AND TARGET-CLASS SCOPES PER §III-E. "—" INDICATES THE TOOL IS NOT LANGUAGE-APPLICABLE.

	VCF			**SAM**		
Tool	htsjdk	vcfpy	noodles	htsjdk	biopython†	seqan3
BIOTEST	32.40±0.71	85.66±0.14	8.10±0.27	21.14±1.42	25.48±0.79	**97.44±0.00**
Jazzer [33]	36.59±1.34	—	—	23.61±1.21	—	—
Atheris [34]	—	87.37±1.83	—	—	58.00±0.36	—
cargo-fuzz [35]	—	—	8.74±1.91	—	—	—
libFuzzer [32]	—	—	—	—	—	90.57±0.00
AFL++ [31]	—	—	—	—	—	73.22±3.57
Pure-Random	0.00±0.00	0.89±0.10	0.00±0.00	1.20±0.06	0.24±0.47	7.19±0.57
BIOTEST engine	PIT	mutmut	cargo-mutants	PIT	mutmut-AST	DIY mull-style
n_{biotest}	10	10	10	10	4	4
n_{baseline}	10	6	9	6	4	3 / 4‡

† biopython matched-budget caps both BIOTEST and Atheris at the first 262 of 523 generated mutants. Full-population BIOTEST score: 25.81±0.00%.
‡ Close-margin pairs were boosted from n=4 to n=10 on BIOTEST, and baseline n is the achieved value shown. Close-margin baseline values are Jazzer/SAM n=6, Atheris/vcfpy n=6, and cargo-fuzz/noodles n=9 from documented run failures. seqan3 baseline values are libFuzzer n=3 and AFL++ n=4.

TABLE II

FINAL LINE COVERAGE MEAN ± SAMPLE STD (%) ACROSS ALL (SUT, FORMAT) PAIRS AND TESTING TOOLS. EACH COLUMN'S TOOLS SHARE THE SAME SCOPE FILTER AND LANGUAGE-NATIVE COVERAGE TOOL. "—" INDICATES THE TOOL IS NOT LANGUAGE-APPLICABLE.

	VCF			**SAM**		
Tool	htsjdk	vcfpy	noodles	htsjdk	biopython	seqan3
BIOTEST	**36.40±6.35**	**67.05±4.30**	**23.85±10.70**	20.85±2.30	48.65±4.25	91.35±0.33
Jazzer [33]	34.69±0.03	—	—	**25.05±0.18**	—	—
Atheris [34]	—	55.01±0.04	—	—	**54.40±0.48**	—
cargo-fuzz [35]	—	—	22.86±0.11	—	—	—
libFuzzer [32]	—	—	—	—	—	**98.45±0.58**
AFL++ [31]	—	—	—	—	—	79.11±0.00
Pure-Random	1.49±0.05	2.72±0.00	0.00±0.00	2.19±0.26	1.84±1.16	78.30±8.07

coverage of any tool we measured (36.40±6.35% on htsjdk, 67.05±4.30% on vcfpy, 23.85±10.70% on noodles). The noodles spread is wide because each fresh BIOTEST rep on a 6,920-line target hits a different control-flow plateau before its stop-criterion fires, but the mean still leads the strongest coverage-guided baseline. On SAM BIOTEST trails the strongest baseline on each parser: by 4.20 pp on htsjdk (20.85% vs Jazzer's 25.05%), by 5.75 pp on biopython (48.65% vs Atheris's 54.40%), and by 7.10 pp on seqan3 (91.35% vs libFuzzer's 98.45%). The pattern is consistent with the mutation-adequacy result of §V-A, and the VCF/SAM asymmetry has a concrete structural explanation. VCF depends on declarative typing constraints in the header (`INFO`, `FORMAT`, contig declarations) that constrain record-level parsing, and BIOTEST's spec-grounded generator exercises these constraints preferentially because each is named in the spec and retrieved by LLM mining. SAM depends more on flag-bit combinations and CIGAR-shape diversity within record bodies, where coverage-guided byte mutators accumulate inputs reaching rare control-flow paths more efficiently than spec-driven enumeration. The same asymmetry surfaces on the mutation-adequacy axis [10], [11].

C. Real-Bug Detection

a) Setup and manifest construction.: We complement §V-A with a real-bug benchmark in the Magma [4] / FuzzBench [39] ground-truth style. The manifest is 32 historical parser bugs (23 VCF +9 SAM), each pinned to a verified $V_{\text{pre}} \rightarrow V_{\text{post}}$ commit pair. Candidates are sourced from each SUT's upstream release notes and `bug`-label PRs over the last five years (we do not mine `git log` for suspicious commits). A candidate enters iff all four hold: (i) reachable through the file-input pipeline as a parse exception, canonical-JSON value diff, or write-roundtrip diff; (ii) pre-fix and post-fix observably differ on the same input; (iii) the anchor pair reproduces empirically; (iv) both versions install as a pinned artefact. Failures are transparently dropped per standard ground-truth bench protocol [4], with the drop log in the artefact. Each entry ships a minimum-shape text PoV authored by reading the fix's `---/+++` diff.

b) Detection predicate, tool slate, and budget.: Detection follows a pre-fix / post-fix differential predicate in the ground-truth fuzzing style of Magma [4]: tool T detects bug B

TABLE III
REAL-BUG DETECTION ON THE 32-BUG MANIFEST (23 VCF + 9 SAM; §V-C). DETECTED: AT LEAST ONE TOOL-PRODUCED TRIGGER SATISFIES SILENCE-ON-FIX. FALSE+: POST-FIX-SURVIVING CRASHES (REAL DEFECTS, NOT THE TARGET BUG).

	paradigm	DETECTED	FALSE+
BIOTEST	file-input differential	$\mathbf{10}/32$	0
Jazzer [33]	in-process crash-only	$0/32$	10
Atheris [34]	in-process crash-only	$0/32$	0
cargo-fuzz [35]	in-process crash-only	$0/32$	0
libFuzzer [32]	in-process crash-only	$0/32$	1
Pure-Random [10]	uninformed floor	$0/32$	0
EvoSuite-anchor[♭]	Java unit-level GA	$4/32$	0
Randoop[♭] [38]	Java unit-level random	$3/32$	0
Any-tool union (file-input ∪ unit-level)		$\mathbf{14}/32$	—

[♭] Unit-level Java baselines exercise method APIs, not file inputs. Their detections are drawn from htsjdk regressions outside the input-fuzzing slate.

iff some input I that T's adapter produced during the budget satisfies $signal_T(I, V_{\text{pre}}) \neq signal_T(I, V_{\text{post}})$. The slate runs eight tools: the five coverage-guided fuzzers of §V-A, BIOTEST, an uninformed Pure-Random floor [10], and two unit-level Java baselines (EvoSuite [40] under an "EvoSuite-anchor" protocol that generates a JUnit suite against V_{post} and replays against V_{pre}, plus Randoop [38]). Per Magma-style methodology [4], every input fuzzer's seed corpus includes the manifest's canonical PoV alongside Tier-1+2 spec-example seeds, equalising starting positions. The per-bug budget is 7200 s, gated by a language-eligibility matrix.

c) Headline.: Table III reports per-(SUT, format) detection counts. BIOTEST detects $10/32$ (htsjdk/VCF 4, htsjdk/SAM 2, vcfpy/VCF 3, noodles/VCF 1), every other input fuzzer detects $0/32$, and the any-tool union with the unit-level Java baselines is $14/32$. Forward (pre-fix rejects, post-fix accepts) fires on eight *over-strict* regressions spanning htsjdk stringency edge-cases, vcfpy lazy-evaluation traps, and a noodles writer round-trip. Reverse (pre-fix wrongly accepts, post-fix rejects) fires on two htsjdk *accept-when-should-reject* regressions where the buggy version silently mis-classified or mis-counted a record. Crash-only oracles surface neither shape without a deliberate exception [10], which is why the crash-only slate scores 0 on every (SUT, format) row.

d) Why the unit-level baselines reach four htsjdk bugs the input fuzzers do not.: EvoSuite [40] synthesises JUnit cases by genetic search over method-call sequences, and Randoop [38] constructs feedback-directed random sequences over the same surface. Both bypass file parsing entirely by constructing `VariantContext` / `SAMRecord` instances via the API. Of the four manifest entries reached this way and not by the input-fuzzing slate, one is a `VariantContextBuilder` regression structurally out-of-scope for any file-input fuzzer, and the other three are parse-time-reachable in principle but BIOTEST's spec-grounded MR transforms in the per-cell budget did not synthesise a trigger that isolated the patch from co-occurring spec violations.

e) Why BIOTEST *leads in absolute count.:* Despite seeing the SUT only through file parse + round-trip, BIOTEST is above both unit-level baselines on absolute count ($10 > 4 > 3$). Two effects dominate. First, BIOTEST applies to all four host languages while EvoSuite and Randoop are Java-only and run on the htsjdk row alone (12 of 32 cells). On that shared subset BIOTEST still leads ($6/12$ vs $4/12$ vs $3/12$). Second, the differential and metamorphic oracles surface silent value-disagreement bugs (e.g. mis-counted AC/AN/AF) that neither crash-only fuzzers nor unit-level GA can detect without an external comparator: crash-only oracles by construction [10], and unit-level GA because its assertions are derived from a single SUT version's observed output.

VI. THREATS TO VALIDITY

a) Construct.: Mutation kills are an imperfect proxy for real-fault detection. Equivalent mutants depress every tool's score equally, and we minimise the asymmetry by reusing standard operator sets [22]. The real-bug benchmark of §V-C grounds adequacy in real bugs as construct triangulation.

b) Internal: bug-bench manifest is post-audit.: The 32-bug manifest reflects a second-pass audit that dropped four SAM entries failing review and added three fresh htsjdk-SAM regressions, with a strict-stringency verification policy added after the initial freeze. Neither choice was blinded to which bugs BIOTEST's paradigm could reach. We apply the post-revision predicate uniformly to every tool to mitigate.

c) Internal: bug-detection attribution.: The bench seed corpus follows Magma-style PoV-seeding [4], equalising starting positions across all input fuzzers. Detection is verified on tool-output triggers under a pre-fix / post-fix differential predicate in the ground-truth fuzzing style of Magma [4]. Canary-based attribution from the same paper or magic-number injection per LAVA [41] would supersede the predicate but require per-bug source instrumentation across all four host languages, which we defer to future work.

d) Internal: rank-level ablation and LLM nondeterminism.: The thirteen-rank corpus stack ships in its deployed configuration with a single end-to-end kill rate per pair. The only rank-level ablation reported is Ranks 12-13 on biopython/SAM. Phase B uses a hosted LLM whose sampling is not bit-deterministic, and the seed-corpus MR validator is not

blinded to downstream performance. We ship the curation log with the artefact as mitigation. The four close-margin mutation-adequacy pairs run at $n{=}10$ and clear family-wise $\alpha{=}0.05$ under Holm–Bonferroni. Biopython and seqan3 remain at $n{=}4$ where samples are deterministic.

e) Internal: ablations not run.: Iteration-count parity on seqan3/SAM would isolate the wall-clock-vs-iteration confound behind the AFL++/libFuzzer gap. A Ranks 1-7 vs 1-13 ablation across every pair would quantify the augmentation-tier contribution beyond biopython/SAM. A hand-rolled differential-only oracle would isolate the $N{\geq}3$ majority-vote contribution.

f) External: bounded generalisation.: The evaluation covers six $(\mathrm{SUT}, \mathrm{format})$ pairs, two textual formats, and four parser languages. The result does not extend to checksum-validated binary formats (BCF, BAM, CRAM) where structural mutations are rejected at I/O time, nor to non-parser SUTs where the round-trip metamorphic relation does not apply.

VII. Conclusion

BioTest is a metamorphic-testing pipeline for textual VCF and SAM parsers whose per-SUT user effort is bounded by what coverage-guided fuzzers already require: a thin parse-and-emit shim and nothing else. The framework combines spec-grounded LLM-driven MR mining, a SUT-agnostic grammar-conforming generator, and a multi-runner consensus oracle applied uniformly across Java, Python, Rust, and C++. We evaluate it under a budget-and-mutation-engine parity protocol against five specialised fuzzers, a Pure-Random floor, and a real-bug benchmark of 32 verified historical parser bugs (§V).

Across the three axes the result is a clean trade-off. BioTest sits within a small single-digit deficit of the strongest coverage-guided baseline on four mutation-adequacy configurations, trails by the margin Klees et al. and Böhme et al. [10], [11] predict qualitatively on byte-content-lenient biopython, and exceeds the strongest baseline on seqan3/SAM through corpus-side retention of typed-exception inputs. On the real-bug benchmark the gap is structurally explained by the dominance of differential-disagreement signals in the manifest's parse-time bug shapes. A natural future direction is a per-SUT coverage-feedback extension on top of the SUT-agnostic core, which would recover some of the biopython gap while preserving the deployment story for users who cannot pay the engineering cost.

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